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PROJECT SYNOPSIS

(Department of ECE)

TITLE :-

Design and Performance Analysis of an FSO Link

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Abstract

Free Space Optical (FSO) communication has emerged as an advanced wireless communication technology capable of delivering fiber-like data rates through optical signal transmission in free space using laser beams. Unlike conventional optical fiber systems, FSO communication eliminates the requirement for physical cables, enabling rapid deployment, flexible network expansion, and cost-effective last-mile connectivity solutions. These advantages make FSO systems highly suitable for applications such as urban communication links, disaster recovery networks, satellite communication, and temporary high-bandwidth connections. However, the performance of FSO links is strongly influenced by atmospheric phenomena including fog, rain, scattering, absorption, and beam divergence, which introduce attenuation and significantly affect link reliability and signal quality [1], [2].

The primary objective of this project is to design and develop an interactive FSO link simulator capable of modeling real-time optical communication behavior under varying environmental and system parameters. The simulator incorporates a physics-based analytical model that calculates link margin by considering atmospheric attenuation losses, geometric spreading losses, transmitter power levels, optical wavelength selection, receiver sensitivity, and propagation distance. By simulating realistic operating conditions, the system generates large datasets representing diverse atmospheric scenarios and link configurations.

To improve prediction accuracy of link margin estimation, machine learning techniques are integrated into the framework. Two models are implemented and evaluated: a conventional Backpropagation (BP) neural network and a Hybrid Newton Raphson Based Optimizer–Backpropagation (NRBO-BP) model. Newton Raphson Optimization enhances neural network training by optimizing weight parameters and improving convergence behavior, thereby overcoming limitations associated with gradient-based learning algorithms [3], [4]. Experimental results demonstrate that the hybrid NRBO-BP approach significantly reduces prediction error, achieving a Mean Squared Error (MSE) of 22.8001 compared to 284.2536 obtained using standard BP, resulting in a performance improvement of approximately 91.98%. The developed simulator therefore provides both an analytical research platform and an intelligent prediction system for evaluating FSO communication performance under realistic environmental conditions.

Motivation

Modern communication infrastructures are experiencing exponential growth due to increasing internet usage, cloud computing, video streaming, and Internet of Things (IoT) applications. Traditional radio frequency (RF) communication systems face several challenges, including limited spectrum availability, interference issues, and regulatory constraints. Although optical fiber communication provides high bandwidth and reliability, installation costs and physical deployment limitations restrict its feasibility in many scenarios such as densely populated cities, remote areas, and temporary communication setups. Free Space Optical communication

presents an attractive alternative by combining the advantages of optical fiber bandwidth with wireless flexibility and rapid installation capabilities [1], [5].

Despite these advantages, practical deployment of FSO systems is hindered by environmental sensitivity. Atmospheric particles cause scattering and absorption of optical signals, resulting in attenuation that varies dynamically with weather conditions. Fog, in particular, causes severe signal degradation due to particle sizes comparable to optical wavelengths, leading to unpredictable link failures [2]. Engineers and researchers therefore require accurate tools capable of predicting link performance before deployment.

Traditional analytical models are often insufficient because FSO link performance depends on multiple interdependent nonlinear parameters. Machine learning methods offer the capability to learn complex relationships directly from data, making them suitable for predicting communication system behavior. However, standard neural networks may suffer from slow convergence and inaccurate weight optimization. This motivates the integration of evolutionary optimization algorithms such as Newton Raphson Based Optimizer to enhance neural network performance [3].

The motivation behind this project is to create a comprehensive simulation environment that not only models physical communication phenomena but also integrates intelligent prediction techniques. The project demonstrates how combining physics-based modeling with computational intelligence can significantly improve system analysis, prediction accuracy, and design efficiency in optical wireless communication systems [4].

Problem Statement

Free Space Optical communication links operate in an open atmospheric channel where environmental conditions introduce significant uncertainty in signal propagation. Parameters such as transmission distance, atmospheric attenuation, wavelength selection, beam divergence, and transmission power interact in a nonlinear manner, making accurate estimation of link margin a complex problem. Link margin determines whether sufficient optical power reaches the receiver to maintain reliable communication; therefore, inaccurate prediction may result in system instability or link failure [2], [6].

Conventional mathematical modeling techniques require simplifying assumptions that may not accurately represent real atmospheric variations. Furthermore, neural network models trained using traditional backpropagation algorithms often converge slowly and may become trapped in local minima during optimization, leading to higher prediction errors and reduced reliability [4]. Another significant challenge is the lack of accessible experimental datasets, as real-world FSO testing requires expensive equipment and controlled environmental conditions.

This project addresses these challenges by developing an FSO simulator capable of generating realistic datasets through a physics-based link margin calculation engine. The simulator models

atmospheric attenuation using predefined weather-dependent coefficients while incorporating geometric spreading loss and system gain analysis to compute link margin dynamically. Generated datasets are then used to train predictive models, and a hybrid NRBO-BP algorithm is applied to optimize neural network parameters globally. The proposed approach improves prediction accuracy while providing a flexible experimental platform for analysing optical wireless communication systems under varying operational conditions [3], [6].

Result Analysis

Model performance evaluation is conducted using Mean Squared Error as the primary metric. Visualization of prediction results shows that the standard BP model produces scattered predictions with noticeable deviation from the ideal prediction line, indicating higher error variance. In contrast, the hybrid NRBO-BP model demonstrates predictions closely aligned with the perfect prediction line, confirming improved generalization capability and accuracy. The reduction in MSE from 284.2536 to 22.8001 highlights the effectiveness of swarm-based optimization in enhancing neural network learning. The hybrid model achieves better parameter optimization, avoids premature convergence, and accurately captures nonlinear relationships between atmospheric conditions and link margin. These results validate the integration of intelligent optimization techniques for communication system performance prediction.

Conclusion

The project successfully demonstrates the design and implementation of an FSO link simulator integrated with intelligent prediction models. By combining physics-based optical communication modeling with hybrid machine learning optimization, the system provides accurate link margin prediction under diverse environmental conditions. The hybrid NRBO-BP model significantly improves prediction performance compared to conventional neural networks, confirming the effectiveness of evolutionary optimization techniques in communication system analysis. The developed simulator serves as a valuable educational, analytical, and research tool for studying optical wireless communication systems and exploring advanced AI-assisted communication design approaches.

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